Laborator 11

15.12.2023 12:14

miercuri, 9:30
13 ian 10:00-13:00 test lab
amf 1,3

V.a. continue

Caz discret $X: \begin{pmatrix} \overset{A}{\boxed{x_1, x_2, \dots x_n}} \\ p_1, p_2, \dots p_n \end{pmatrix}$

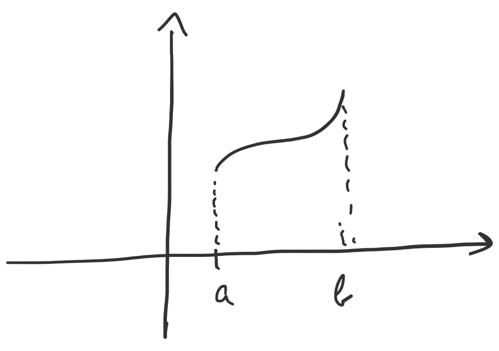
Caz continuu: $X: \begin{pmatrix} x \\ f(x) \end{pmatrix} \quad x \in \mathbb{R}$

nu este prob!
se n. funcție densitate de prob

$$\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}$$
$q \in (-1,1)$

ex $q = \frac{1}{2}$ $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n = 1$

$$\mathbb{P}(a < X < b) = \int_a^b f(x)\,dx$$



f este funcție de densitate

$$\begin{cases} f(x) \geq 0 \quad \forall x \in \mathbb{R} \\ \int_{-\infty}^{\infty} f(x)\,dx = 1 \end{cases}$$

ex $f(x) = \begin{cases} c\cdot(x^2+2), & x \in [0,1] \\ 0, & \text{în rest} \end{cases} = c(x^2+2)\cdot \mathbb{1}_{[0,1]}(x)$

$$\boxed{\text{supp}(f) = \{x \in \text{Dom}(f),\ f(x) \neq 0\}}$$

$$\mathbb{1}_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$

$$E(g(x)) = \int_{-\infty}^{\infty} g(x)\cdot f(x)\,dx$$

$$\mathbb{E}(x^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx$$

① $f(x) = \begin{cases} c\cdot(4x-2x^2), & 0 < x < 2, \quad c \in \mathbb{R} \\ 0, & \text{în rest} \end{cases}$

a) f densitate de prob $\iff \begin{cases} f(x) \geq 0 \quad \forall x \geq 0, \forall x \\ \int_{-\infty}^{\infty} f(x)\,dx = 1 \end{cases}$

1) $f(x) \geq 0 \iff c\cdot \underbrace{2x}_{>0} \underbrace{(2-x)}_{>0} \geq 0 \quad \forall x \in (0,2) \Rightarrow c > 0$

2) $\int_{-\infty}^{\infty} f(x)\,dx = 1 \iff \int_0^2 c\cdot(4x-2x^2)\,dx = 1 \iff c\left(4\int_0^2 x\,dx - 2\int_0^2 x^2\,dx\right) = 1$

$\iff c\cdot\left(4\cdot\left(\frac{x^2}{2}\Big|_0^2\right) - 2\cdot\frac{x^3}{3}\Big|_0^2\right) = 1$

$\iff c\left(4\cdot\frac{4}{2} - 2\cdot\frac{8}{3}\right) = 1$

$\iff c\left(8 - \frac{16}{3}\right) = 1$

$\iff c\cdot\frac{8}{3} = 1$

$\iff c = \frac{3}{8}$

b) $\mathbb{P}(X > 1) = \int_1^{\infty} f(x)\,dx = \int_1^2 \frac{3}{8}(4x-2x^2)\,dx = \frac{3}{8}\cdot\left(4\int_1^2 x\,dx - 2\int_1^2 x^2\,dx\right)$

$= \frac{3}{8}\left(4\cdot\frac{x^2}{2}\Big|_1^2 - 2\cdot\frac{x^3}{3}\Big|_1^2\right) = \frac{3}{8}\left(4\cdot\left(\frac{4}{2} - \frac{1}{2}\right) - 2\cdot\left(\frac{8}{3} - \frac{1}{3}\right)\right)$

$= \frac{3}{8}\left(4\cdot\frac{3}{2} - 2\cdot\frac{7}{3}\right) = \frac{9}{4} - \frac{7}{4} = \frac{2}{4} = \frac{1}{2}$

c) $F(x) = \int_{-\infty}^{x} f(t)\,dt$

$$f(t) = \begin{cases} 0, & t \leq 0 \\ \frac{3}{8}(4t-2t^2), & t \in (0,2) \\ 0, & t \geq 2 \end{cases}$$

I $x \leq 0$

$F(x) = 0$

II $x \in (0,2)$

$F(x) = \int_{-\infty}^{0} 0\,dt + \int_0^x \frac{3}{8}(4t-2t^2)\,dt$

$= \frac{3}{8}\left(4\frac{t^2}{2}\Big|_0^x - 2\frac{t^3}{3}\Big|_0^x\right)$

$= \frac{3}{8}\left(4\frac{x^2}{2} - 2\cdot\frac{x^3}{3}\right)$

$= \frac{3}{8}\left(2x^2 - \frac{2}{3}x^3\right)$

III $x > 2$

$F(x) = \int_{-\infty}^{0} 0\,dt + \int_0^2 \frac{3}{8}(4t-2t^2) + \int_2^{\infty} 0\,dt = 1$

$$F(x) = \begin{cases} 0, & x \leq 0 \\ \frac{3}{8}\left(2x^2 - \frac{2}{3}x^3\right), & x \in (0,2) \\ 1, & x \geq 2 \end{cases}$$

$\mathbb{P}(X > 1) = 1 - \mathbb{P}(X \leq 1) = 1 - F(1) = 1 - \frac{3}{8}\left(2 - \frac{2}{3}\right) = 1 - \frac{3}{4} + \frac{1}{4} = \frac{1}{2}$

T: $\mathbb{E}(X)$, $\text{Var}(X)$

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$$

$$\mathbb{E}(x^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx$$